



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

CDF shall be presented initially in all academic fora (BoS, BoF, AC). Subsequent revisions shall require approval from the respective BoS only.

Course Information

Course Code: MTH262	Course Title: Statistics and Probability Theory
Credit Hours: 3(3,0)	Lecture Hours/Week: 3
Lab Hours/Week: 0	Pre-Requisites: None

Course Objective

- Apply statistical techniques to analyze data, calculate descriptive statistics, and interpret results for decision-making.
- Learn to use probability theory to model and solve real-world problems involving uncertainty and randomness.
- Gain proficiency in hypothesis testing and confidence intervals to draw meaningful conclusions from data.
- Understand and apply regression analysis and correlation techniques to explore relationships between variables and make predictions.

Catalogue Description:

This course covers essential statistical techniques for analyzing data, calculating descriptive statistics, and interpreting results for decision-making. It includes probability theory for modeling uncertainty and solving real-world problems. Key topics include hypothesis testing, confidence intervals, and drawing meaningful conclusions from data. Additionally, the course focuses on regression analysis and correlation techniques to explore variable relationships and make predictions.

Unit wise Major Topics:

Unit	Topic	No. of Teaching Hours
1.	Introduction to Statistics, Basic definitions, Frequency distribution, Graphs.	5

2.	Measures of central tendency and measures of dispersion. Properties of variance and applications of standard deviation. Quartiles and percentiles. Box and Whisker plot	9
3	Set theory, Basics of probability theory, laws of probability, Bays Rule.	9
4	Random variable, Probability mass and density functions, Cumulative distribution functions, Expectation of Random Variable, Variance of Random variable	7
5	Discrete probability distributions and continuous probability distributions	6
6	Confidence Interval, Testing of hypotheses	6
7	Simple linear regression and Correlation	3

Total Contact Hours	45
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Mapping of CLOs and PLO/Gas

Sr.#	Course Learning Outcomes	Blooms Taxonomy Learning Level	PLO/GA
CLO-1	Describe and understand methods for data collection, organization, presentation, and descriptive statistical analysis using appropriate numerical measures and graphical techniques.	C2	PLO-
CLO-2	Apply fundamental concepts of probability theory to model uncertainty and solve real-world and data-driven problems.	C3	PLO-
CLO-3	Analyze and evaluate statistical inference problems by estimating parameters and testing hypotheses using standard techniques such as Z-tests, t-tests, F-tests, and chi-square tests.	C4	PLO-
CLO-4	Apply statistical software such as R, Python, or SPSS to perform data analysis, generate visualizations, and interpret statistical outputs to support data-driven decision making.	C5	PLO-

CLO Assessment Mechanism				
Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4
Quizzes	✓	✓	✓	✓
Assignments	✓	✓	✓	✓
Mid Term Exam	✓	✓	✓	
Final Term Exam		✓	✓	✓
Text and Reference Books				
Textbook: 1. Introductory Statistics, 10th Edition, Neil A. Weiss, Pearson, 2016. Reference Books: 2. Practical Statistics for Data Scientists, 2nd Edition, Peter Bruce, Andrew Bruce, & Peter Gedeck, O'Reilly, 2020. 3. Probability and Statistics for Computer Science, 1st Edition, David Forsyth, Springer, 2018.				